Problem-3: Part-C

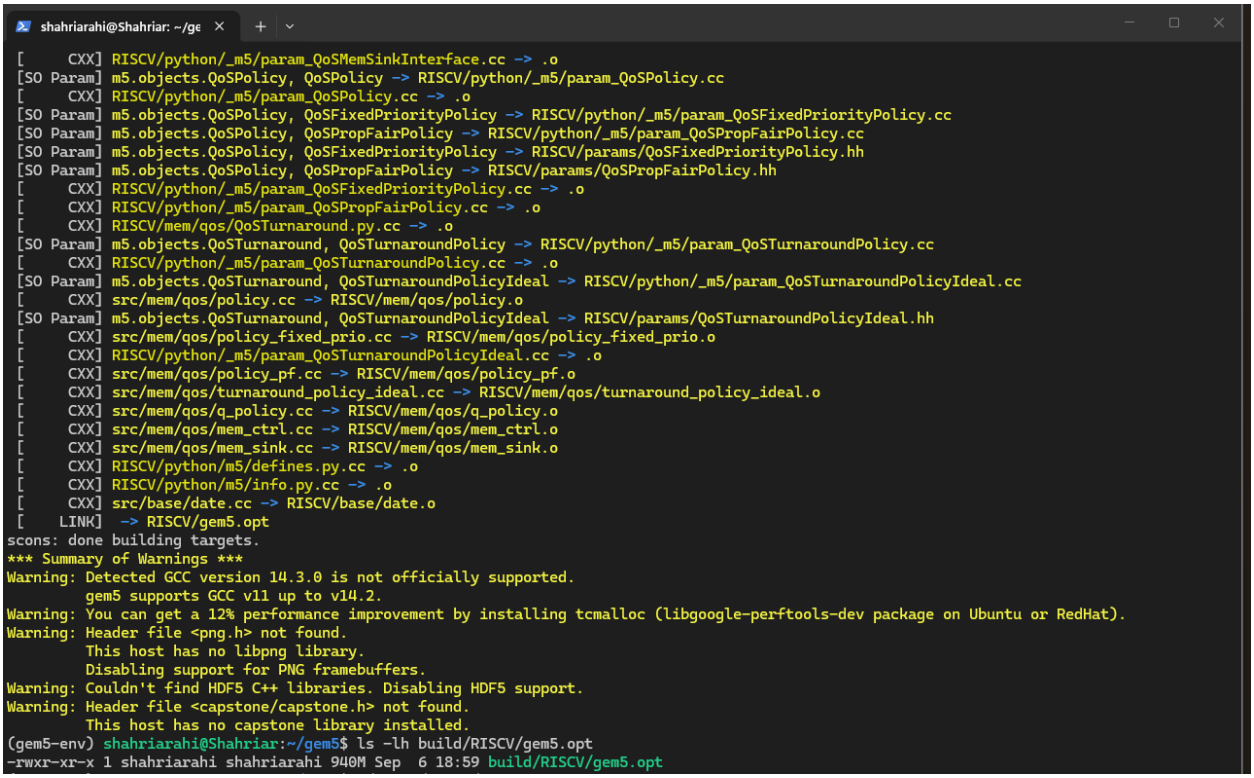
The bottleneck is that 35% of the baseline execution time cannot be accelerated by the NPU. This could include data transfers, CPU control instructions, synchronization, or non-matrix computations. Even if NPU's operations take zero time, the application must still complete the unenhanced portion's work in the normal way. Therefore, acceleration alone cannot exceed the speedup ceiling.

This explains why architects prioritize **"making the common case fast"**: an optimization's value depends on how much of the application's total execution time it saves. A spectacular speedup of a rarely used operation can have very little effect on the whole program. For example, accelerating an operation that occupies only 1% of runtime yields only about a 1% overall performance improvement. A complex hardware unit may therefore consume substantial area, power, and design effort while delivering little practical benefit.

Part-B : Hands-On gem5 Simulation Exercise

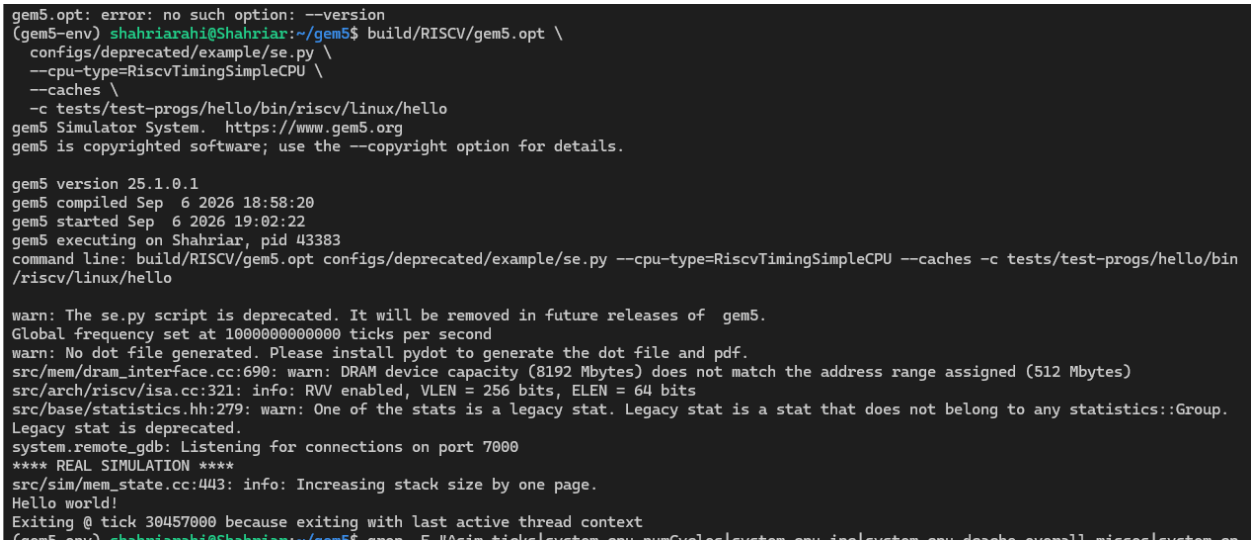
Step-1: SCons Simulator Compilation

The SCon build has been conducted successfully.



Step-2: Simple Executable Run in SE Mode

I have used the commands from the assignment and generated a “Hello World” line as shown below.



Step-3: Navigating Telemetry & Metric Extraction

Gem5 Stat Key	Physical Hardware Metric	Extracted Value
Sim_ticks	Total simulated benchmark runtime in picoseconds (ps). (Include your conversion to nanoseconds (ns)).	30,457,000
System.cpu.numCycles	Total simulated processor clock cycles spent processing the dynamic instruction stream.	60,914
System.cpu.ipc	Instructions executed per clock cycle. (Compute its reciprocal to find the effective Cycles Per Instruction, CPI).	0.090587

System.cpu.dcache.overall_misses	Total cache lookup misses recorded on the L1 Data Cache interface during execution.	135 misses
System.cpu.dcache.overall_miss_rate	The fraction of L1 Data Cache references that resulted in misses, causing memory hierarchy stalls.	0.065217

Step-4: Reflection

Hello World took 60,914 cycles to retire its instruction stream, an IPC of 0.0906 (≈11 CPI). For a workload this small, that figure says more about startup cost than steady-state behavior: the instruction count is dominated by process setup and a single write syscall, so the pipeline has little opportunity to fill. The 135 L1 data cache misses (6.52%) are consistent with this — cold-start compulsory misses on a working set that is never revisited.